# DISPOSITION FORM

SECURITY CLASSIFICATION (If any)

29

| FILE NO. Form 14 No. 53-11-23-5 | SUBJECT Informal Report of Special Investigation of Major Aircraft Accident Involving F-89C, Serial No. 51-5853A, over Lake Superior, on 23 November 1953 |
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TO AFCFS-1A FROM AFCFS-2A DATE 6 Jan 54 COMMENT NO. 1

Maj Watts/fj/5131/8-802

FACT ACCIDENT

1. F-89C, Serial No. 51-5853A, assigned to the 433rd Fighter-Interceptor Squadron, Truax Field, Wisconsin, was reported missing over Lake Superior at approximately 2000 Eastern Standard Time (EST) on 23 November 1953. The aircraft was scrambled from Kinross Air Force Base, Michigan, to participate in an Active Air Defense Mission. The aircraft and aircrew had not been located as of 1 January 1954.

CONCLUSIONS

2. It is concluded that:

a. The aircraft probably crashed in the Canadian waters of Lake Superior just prior to or at the time of interception (see paragraphs 4 and 5).

b. The Ground Control Intercept (GCI) station had been vectoring the aircraft to intercept an unidentified aircraft (see paragraphs 4 and 6).

c. The GCI station maintained radar contact with both the interceptor and the unidentified aircraft until the time of interception (see paragraphs 4 and 5).

d. Interception techniques, communications, aircraft maintenance or pilot's experience apparently were not cause factors in the accident (see paragraphs 4, 5, 8, and 9).

e. Instrument flying conditions prevailed during the time of interception and could be a cause factor of the accident (see paragraph 7).

ACTION TAKEN

3. None.

HISTORY OF FLIGHT

4. On 23 November 1953, F-89C, Serial No. 51-5853A was scrambled by "Naples" GCI to intercept and identify an unknown aircraft flying over Lake Superior. The interceptor became airborne from Kinross Air Force Base, Michigan at 1822 EST. Original radar control of the aircraft was maintained by "Naples" GCI and at 1841 EST control was transferred to "Pillow" GCI. The aircraft was flying at 30,000 feet at this time. At 1847 EST, at the request of "Pillow" the aircraft descended to 7,000 feet to begin the interception. Location of the aircraft was then approximately 150 miles northeast from Kinross AFB and over northern Lake Superior. At 1851 EST, the

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interceptor pilot was requested to turn to a heading of 20 degrees to the cut-off vector. After the turn was completed, the pilot was advised the unidentified aircraft was at 11 o'clock, ten miles distant. Radar returns from both aircraft were then seen to merge on "Pillow's" radar scope. The radar return from the other aircraft indicated it was continuing on its original flight path, while the return from the F-89 disappeared from the GCI station's radar scope.

INVESTIGATION AND ANALYSIS

5. Radar reception was considered exceptionally good during the entire period of the interceptor's flight. The GCI station was receiving the F-89's radar return plus the F-89's Identification, Friend or Foe (IFF) return. At the time of interception, when the target and the interceptor radar returns merged, the GCI controller first assumed that the F-89 was flying formation with the other aircraft. This is normal procedure when identifying an unknown aircraft. The simultaneous disappearance of the F-89's radar and IFF return indicates the accident occurred at this time. If the IFF had suddenly become inoperative then the regular radar return would continue to be observed by the GCI operator. Both "Naples" and "Pillow" GCI stations reported difficulty in receiving radio transmissions from the pilot of the F-89 during the initial period of flight. At one time, the pilot asked "Pillow" GCI if the intercept should be discontinued because of the weak reception at "Pillow". He was advised to continue at his own discretion. This radio difficulty was apparently temporary and probably due to weather conditions. Later when the pilot was told to descend to the unknown aircraft's altitude, the radio transmission was received loud and clear by "Pillow" GCI.

6. The unknown aircraft being intercepted was a Royal Canadian Air Force Dakota (C-47), Serial No. VC-912, flying from Winnipeg to Sudberry, Canada. At the time of interception it was crossing Northern Lake Superior from west to east at 7,000 feet. This flight path was approximately 30 miles south of the intended flight path. The pilot stated that he was on top of a 5,000 foot undercast and at the approximate time of interception he was flying in the clear and visibility was unlimited. He also stated he did not know he was being intercepted and that he did not see the F-89.

7. Weather conditions during the period of the interceptor's flight were marginal and apparently deteriorated immediately after the time of interception. The Canadian pilot of the aircraft being intercepted stated that, although he was in the clear at the time of the interception, just prior to this he had been flying in instrument conditions. Other pilots who took off to search for the missing F-89 reported various weather conditions but an analysis of all reports indicates definite instrument conditions with snow showers in and around the Lake Superior area. Rime icing had been forecast for this area and was also reported throughout this general area.

8. This F-89 was equipped with J35-A-41 engines which incorporates retractable inlet screens. This would eliminate engine screen icing as a possible cause factor. The aircraft had been previously flown for one hour on the day of the accident and the pilot reported no discrepancies after the flight. The aircraft was then serviced and returned to alert status. Technical Orders not complied with on the aircraft are not considered as a cause factor in this accident. The aircraft weight and balance was within limits at time of takeoff.

9. The pilot and radar observer were assigned to the 433rd Fighter Interceptor Squadron, Truax AFB, Wisconsin. They were on temporary duty at Kinross AFB, Michigan, while the base's regularly assigned personnel were firing gunnery at Yuma, Arizona. The pilot had a total of 811:00 hours of which 121:40 hours were in F-89 type aircraft. He had 101:00 instrument hours and 91:50 hours night time. The radar observer was relatively inexperienced with a total of 208:45 hours of which 11:30 hours were at night. The medical history of the pilot and radar observer is not considered as pertinent to the accident.

10. Search for the missing aircraft was conducted by both USAF and RCAF aircraft without success. Although 80 per cent area coverage was reported, heavy snows precluded effective land search. All civilian reports of seeing or hearing the aircraft were investigated with negative results.

SUBSTANTIATING DATA ON FILE IN THE DIRECTORATE OF FLIGHT SAFETY RESEARCH

11. The following substantiating data pertaining to special investigation of aircraft accident involving F-89C, Serial No. 51-5853A, are on file in the Directorate of Flight Safety Research and can be obtained upon request:

a. Orders directing the investigation

b. List of personnel participating in the investigation

c. Statistical data

d. Missing Aircraft Report, Preliminary Report, Search Discontinuance Report

e. Form 14A, Pilot's Flight Records, Radar Observer's Flight Records

f. Pilot's activities prior to flight

g. Scramble Clearance, Weather

h. Statements, Ground Controller's Report

i. Accident Board Proceedings

j. Area map with Bogey's and Interceptor's positions plotted

k. Aircraft Engineering Data

JAMES I. CONNETT, Colonel, U. S. Air Force
Chief, Inves & Safety Engr Division

COORDINATION
Watts
Anderson
Alexander

S T A T E M E N T

When A-27-T was picked up by Pillow (P-16) it was believed to be VC-912, but because the aircraft was off the flight plan course by about 30 miles, it was classified as "Unknown". Word was received from Naples (P-66) that Horsefly wanted a correlation check on the track.

At 2317Z, Naples scrambled Avenger Red on a vector of 300° angels 30. When the aircraft became airborne at 2322Z the pilot gave Naples a call on channel 10 (UHF-2). Because their reception of Avenger Red was weak, they attempted to pass control about 2335Z. At that time, we could read Avenger Red only very faintly. Both "Receivers" and "Transmitters" at this station were notified of the difficulty encountered in reading Avenger Red, and checks of our equipment were made with no apparent malfunction of equipment.

At 2341Z control of Avenger Red was passed to this station, although the UHF reception at this station was poor. Avenger Red was, however reading our Transmissions about 4x4. Avenger Red was then directed to switch to Channel 12 (UHF-1), and upon so doing it was learned that reception was no better, whereupon both the fighter and this station returned to Channel 10.

About 2345Z Avenger Red asked if the intercept should be discontinued because of the weak UHF reception at this station. The pilot was then informed that the continuance of the mission was up to his descretion.

About 2347Z Avenger Red was given a Port turn to 270 and was directed to descend to angels 7. When Avenger Red answered with "Steady on 270", he came in loud and clear at this station, and was informed of such. Shortly thereafter, a clock and range report was given the pilot along with the

bogey's track and altitude. This information was acknowledged and the pilot informed us that he was passing through angels 25. (Approx. 23472)

About this point, Avenger Red was informed that he might encounter icing conditions on the descent to the bogey's altitude, such information having been received from Horsefly. This transmission was acknowledged by Avenger Red, who then asked for "Pigeons to Home Plate". Pigeons to Kinross were given as 150 - 125 miles, which was acknowledged by the pilot.

At Approximately 2351Z Avenger Red was given a Starboard turn to 020 deg., which was the cutoff vector. About 2352Z Avenger Red was informed that the bogey was at 11:00-10 miles, which was acknowledged by the pilot. Another clock and range report was given, allong with the information that the bogey was passing from port to starboard, but no acknowledgement was received for either transmission. Just before the target merged on the PPI Scope, Avenger Red was informed that he would be set up for another pass on the bogey if he had not received a contact. When no answer was received from Avenger Red, it was believed that his Radar Observer had established contact with the bogey on his weapon and had taken over the intercept. Had such been the case, the Radar Observer would likely have been keeping the pilot very busy — too busy to acknowledge my transmissions. Thus, when there was no separation of targets on the PPI Scope after the fighter had merged with the bogey, it was assumed for a couple of minutes that Avenger Red was flying formation with the bogey. Every few seconds, however, an attempt was made to contact Avenger Red. When no contact could be established, Naples was contacted and advised to try contacting Avenger Red. Naples efforts to contact the fighter were unsuccessful also.

Shortly after the two blips merged on the PPI Scope, the IFF return disappeared. It should be noted also that the fighter had been painting a radar return on the scope in addition to the IFF return. Thus, it is strange that no radar return on the fighter was picked up after the merger, had the IFF suddenly become inoperative.

At the direction of Horsefly, two additional fighters were scrambled, (Avenger Black and Avenger Purple) to proceed to the area of Avenger Red's disappearance. At Naple's direction, Avenger Black gav e several calls to Avenger Red with the hope that contact might be established between the two aircraft. When all attempts to contact Avenger Red proved futile Avenger Black and Avenger Purple were vectored back to base by Naples.

When it became certain that the fighter was down, the Coast Guard at Hancock was notified. They in turn called the District Headquarters at Marquette to send out a Cutter from that station. Marquette then called Houghton Radio to confirm the fact a fighter was down, and upon receiving an affirmative answer, dispatched a Cutter for the area in which Avenger Red is believed to be down.

/s/ DOUGLAS A. STUART
2nd Lt., USAF
Controller

A CERTIFIED TRUE COPY

David C. Collins

DAVID C. COLLINS
Captain, USAF
Aircraft Accident Investigating Officer

23 November 1953 A/C NO 51-5853A Lt Moncla

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# S T A T E M E N T

I talked to the pilot of the aircraft which was being intercepted by the F-89 #51-5853A at the time of the disappearance of the F-89. The Aircraft being intercepted was a Royal Canadian Air Force Dakota (C-47) on a flight from Winnipeg to Sudberry Canada. The pilot stated that the weather at the approximate last known position of the F-89 was an undercast with tops at 5000 ft. There was a scattered condition at 10,000 ft and above that the ceiling and visibility was unlimited. The pilot stated that he was flying at 7000 ft in the clear and visibility was unlimited. He stated that he had been in clouds during the earlier portion of his flight and that he had encountered moderate rime icing in the clouds.

The pilot stated that he did not know he was being intercepted and had not seen nor heard of the F-89. He stated that he heard no radio transmissions which could have emanated from the F-89. The pilot could offer no further information to throw any light on the investigation.

David C Collins

DAVID C. COLLINS
Captain, USAF
Aircraft Accident Investigating Officer